

Recall sample cov. for $(U_1, V_1), \dots, (U_m, V_m)$

$$\frac{1}{m} \sum_{i=1}^m (U_i - \bar{U})(V_i - \bar{V}) \quad (*)$$

Now for stationary time series

$$X_1, X_2, \dots, X_n$$

The pairs corresponding to lag $h \in \mathbb{Z}_+$.

are $(X_{1+h}, X_1), (X_{2+h}, X_2), \dots, (X_n, X_{n-h})$
 $U_1 \quad V_1 \quad U_2 \quad V_2 \quad U_m, V_m \quad m=n-h$

Plugging into (*)

$$\Rightarrow \frac{1}{n-h} \sum_{i=1}^{n-h} \left[\underbrace{X_{i+h}}_{U_i} - \underbrace{\frac{1}{n-h} (X_{1+h} + \dots + X_n)}_{\bar{U}} \right] \times \left[\underbrace{X_i}_{V_i} - \underbrace{\frac{1}{n-h} (X_1 + \dots + X_{n-h})}_{\bar{V}} \right]$$

$\bar{U} = \frac{1}{n-h} (X_{1+h} + \dots + X_n)$
 $\bar{V} = \frac{1}{n-h} (X_1 + \dots + X_{n-h})$

Q: Are $\bar{U}$ & $\bar{V}$ the same?

A: No.

Issue: $\bar{U}$ skips $X_1, X_2, \dots, X_h$

$\bar{V}$ skips $X_{n-h+1}, X_{n-h+2}, \dots, X_n$

$$* \hat{\gamma}(h) = \frac{1}{n-h} \sum_{t=1}^{n-h} (X_{t+h} - \bar{x})(X_t - \bar{x})$$

is non-negative definite.

$$\begin{aligned} & \rightarrow \begin{bmatrix} \hat{\gamma}(0) & \hat{\gamma}(1) & \cdots & \hat{\gamma}(m) \\ \hat{\gamma}(1) & \hat{\gamma}(0) & & \vdots \\ \vdots & \hat{\gamma}(1) & \ddots & \vdots \\ \hat{\gamma}(m) & \hat{\gamma}(m-1) & \cdots & \hat{\gamma}(0) \end{bmatrix} = \left[\hat{\gamma}(i-j) \right]_{i,j} \\ & \quad \quad \quad (\hat{\gamma}(t) = \hat{\gamma}(-t)) \\ & \text{hold for } \gamma(h) \quad 0 \leq m \leq n-1 \end{aligned}$$